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Paleomagnetism of Late Mesozoic Granitic Intrusions in Hong Kong: Implications for Upper Cretaceous Reference Pole of South China

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Two Late Jurassic-early Cretaceous granite plutons in Hong Kong were sampled at 16 sites for paleomagnetic determination. The two granites show differences in magnetic properties and mineralogy but similar natural remanent magnetization (NRM) directions. Multiple magnetizations and dual polarity have been detected in the samples. Isothermal remanent magnetism, magnetic susceptibility, and coercivity measurements reveal multidomain titanomagnetite as the principal magnetic carrier. A rating of paleomagnetic results from individual sites was based on the number of specimens, efficacy of the demagnetization procedures, and within-site variance of NRM directions. The pole position from 12 high quality sample sites is 171.9°E, 78.2°N, $A_{95}=10.6°$ The dual polarity and cooling ages of the granites suggest a Late Cretaceous acquisition of the remanent magnetism. Based on four different studies, the Upper Cretaceous reference pole of South China is located at 194.3° longitude, 73.5°N latitude, $A_{95}=10.4°$. Paleomagnetic results from North China, South China, and Eurasia show no large-scale Cenozoic relative movements among the three blocks. But the small separation between the paleomagnetic poles of South China and Eurasia probably implies a minor relative movement since the Late Cretaceous

INTRODUCTION

East Asia consists of 10 to 12 microplates separated by orogenic zones. Collisions among these microplates began during the Paleozoic, with final amalgamation of the plates during the Mesozoic. The paleotectonic history of East Asia can be reconstructed from paleomagnetic, stratigraphic, geochronological, and structural studies of the microplates and their boundaries. Paleontological and paleomagnetic evidence shows that these microplates, including India, Tibet, Indochina, Shan-Thai Plateau, and South China, probably originated as parts of Gondwanaland in the southern hemisphere during the late Paleozoic [e.g., *Ridd*, 1971; *Acharyya*, 1978, 1979]. These continental blocks detached from Gondwanaland, migrated northward, and accreted onto the southern part of Eurasia [*Irving*, 1977; *McElhinny et al.*, 1981; *Chan et al.*, 1984; *Opdyke et al.*, 1986].

The South China block is separated from North China by the Qinling orogenic belt, from Tibetan Plateau in the west by the Lungmenshan thrust, and from the Indosinian block in the southwest by the Red River fault (inset, Figure 1). Many of these tectonic boundaries underwent multiple orogenic movements, and the ages of the continental accretions are controversial. Syntectonic granites and glaucophane schists in the Qinling Mountains have yielded Devonian to Late Carboniferous radiometric ages, indicating a late Paleozoic collision between South China and North China [*Xiao et al.*, 1978, 1980; *Zhang et al.*, 1984]. *Li and Ye* [1980], *Klimetz* [1983], *Zhao and Coe* [1987], and *Yang et al.* [1987], however, subscribe to an alternative hypothesis of a Mesozoic suture of the two blocks. This controversy over the age of the collision might be resolved by invoking a paleozoic back-arc basin or, alternatively, a double closure of North and South China blocks [*Huang*, 1978]. Nevertheless, paleomagnetic results from South and North China are essential for determining the precise tectonic history of the two blocks.

Despite problems of weak rock magnetism, remagnetization, and insufficient information on the structural geology of sample sites, the studies conducted by *McElhinny et al.* [1981], *Chan et al.* [1984], *Lin et al.* [1985], *Huang et al.* [1986], *Opdyke et al.* [1986], and *Zhao and Coe* [1987, 1989] yielded compatible results on the Permian and Triassic paleopoles of South China and indicated a post-Triassic accretion of North China and South China. *Lin et al.* [1985], *Kent et al.* [1986], and *Chan and Zich* [1986] have attempted to determine the Late Cretaceous paleopole of South China. *Courtillot and Besse* [1986] reported a Late Jurassic-early Cretaceous paleomagnetic pole of South China. Although the Triassic and Lower Cretaceous paleopoles of South China are reasonably well determined, the post-Triassic polar wander curve remains poorly constrained.

This study represents an attempt to refine the polar wander path of South China. Two Late Jurassic-early Cretaceous granites in Hong Kong were sampled at 16 sites in 1986 and 1987. Measurements of natural remanent magnetism (NRM) intensity and magnetic susceptibility, alternating-field (AF) and thermal demagnetization, and isothermal remanent magnetism (IRM) experiments were conducted. The paleomagnetic pole obtained in this study is combined with previous results from eastern China [*Kent et al.*, 1986], South Korea [*Otofuji et al.*, 1983] and Guangdong Province [*Hsu*, 1987] to determine a reference pole for South China.

IGNEOUS GEOLOGY OF HONG KONG

The territory of Hong Kong is located in the northeast striking Cathaysian orogenic belt which represents a Mesozoic volcanic arc along the southeastern margin of the Yangtze Platform (Figure 1). In the sample area, Cambrian (?) through Jurassic marine formations are cut by granitic plutons and overlain unconformably by Jurassic to Lower Cretaceous continental sediments [*Lee*, 1985]. In the coastal areas of South China, Yanshanian orogenic activity culminated with syntectonic granite intrusion in three stages over an interval of 60 m.y. during the Late Jurassic through early Tertiary (Table 1) [*Huang*, 1960; *Davis*, 1953; *Allen and Stephens*, 1971]. The first two intrusive stages (phases 1-4) are represented by Late Jurassic granitic batholiths and stocks, and the last stage (phase 5) consists of Late Cretaceous-early Tertiary dike swarms.

Allen and Stephens [1971] distinguished eight granitic bodies in Hong Kong. The most extensively exposed plutons are the Hong Kong granite and the Cheung Chau granite (Figure 1).

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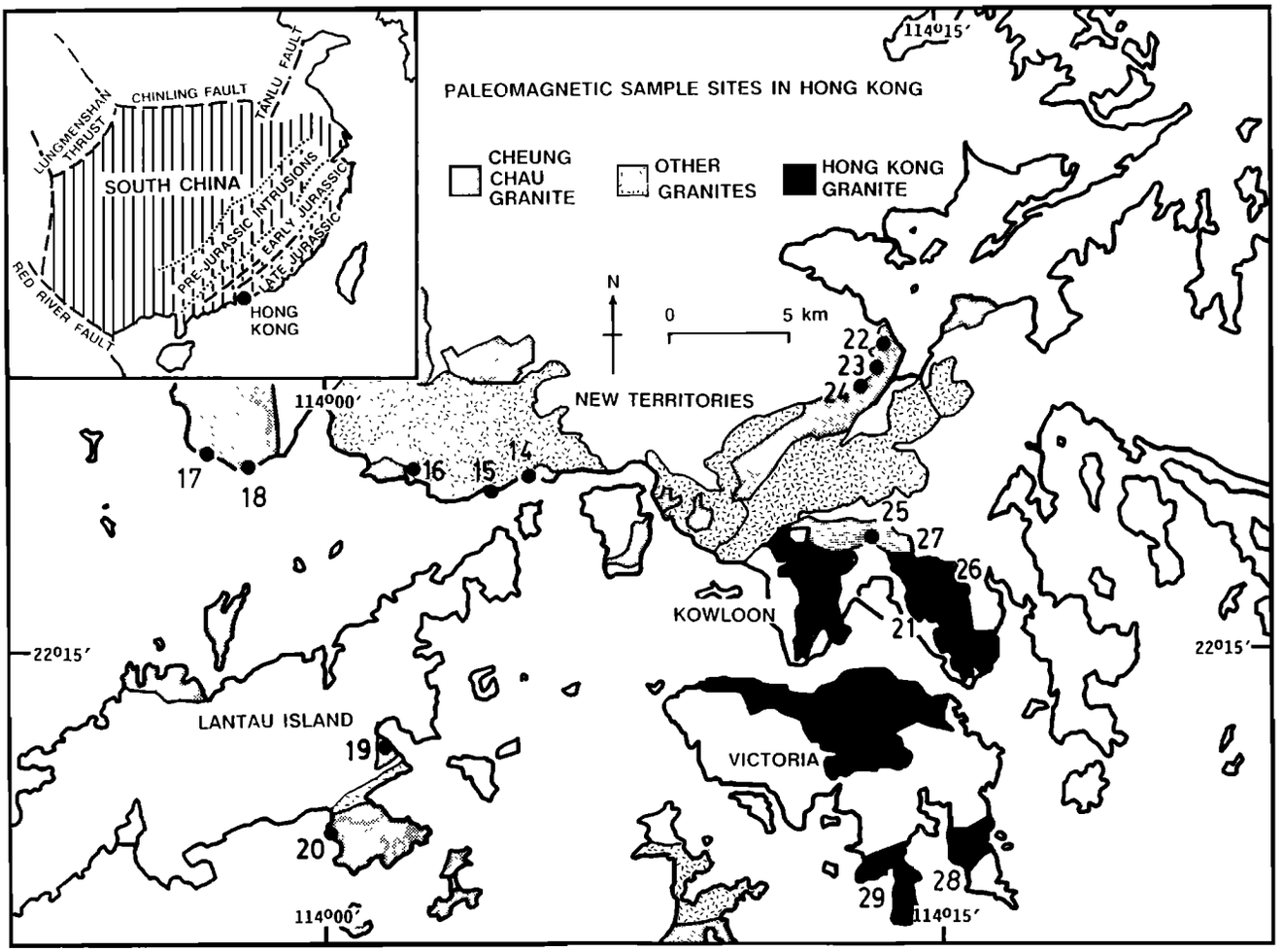


Fig. 1. Paleomagnetic sample sites in Hong Kong. The inset shows the tectonic boundaries of the South China Block and the approximate extent of intrusions of different ages.

These two granites are similar lithologically; they are both pale pink to grey, medium-grained, equigranular, and non-porphyritic, containing 30-40% quartz and carrying microperthite as the principal potassium feldspar and pyrite, epidote, and euhedral magnetites as the accessory minerals. The Cheung Chau granite is considered to be older than the Hong Kong granite because it is intruded by quartz-porphyry dikes while the Hong Kong granite is not [*Allen and Stephens*, 1971]. The presence or absence of porphyry dikes is used for distinguishing between the two granites in the field.

The ages of the intrusive rocks in Hong Kong have been determined both stratigraphically and radiometrically. The absence of contact metamorphism in the Upper Cretaceous continental sediments exposed in the territory provides a minimum age estimate for the granites. A geochronological study of the intrusive rocks from Hong Kong has yielded an age of 160±35 Ma with a $^{87}Sr/^{86}Sr$ ratio of 0.7098±0.0032 for the Cheung Chau granite and an age of 140±7 Ma with a $^{87}Sr/^{86}Sr$ of 0.7027±0.0030 for the Hong Kong granite [*Chandy and Snelling*, 1971]. The results are consistent with the ages of the intrusive rocks in southeastern China obtained by the *Guiyang Institute of Geochemical Research* [1979]. Potassium-argon dating of biotite has yielded cooling ages of 117 Ma for the Hong Kong granite and 135 Ma for the Cheung Chau granite, suggesting a cooling interval of about 25 m.y. for the plutons [*Chandy and Snelling*, 1971].

TABLE 1. Igneous Intrusive Phases in Hong Kong [*Allen and Stephens*, 1971]

Phase	Rb/Sr Age m.y.	K/Ar Age m.y.	Activities
1		134±2	granodiorite
2-3	163±35	137±6	granites and quartz monzonite, including Cheung Chau granite
4	140±7	117±3	Hong Kong granite
5	65±8		dolerite dikes

SAMPLE SITES AND EXPERIMENTAL PROCEDURES

The two granites were sampled at 16 different sites, designated HK14 through HK29, in road cuts and quarries (Figure 1). Three to five block samples were collected at each site; at least four 2.4 cm length by 2 cm diameter core specimens were prepared from each sample.

The experimental procedures included measurements of NRM intensity and magnetic susceptibility, alternating field (AF) demagnetization up to 800 mT, stepwise thermal demagnetization up to 650°C, and IRM acquisition experiments. The NRM

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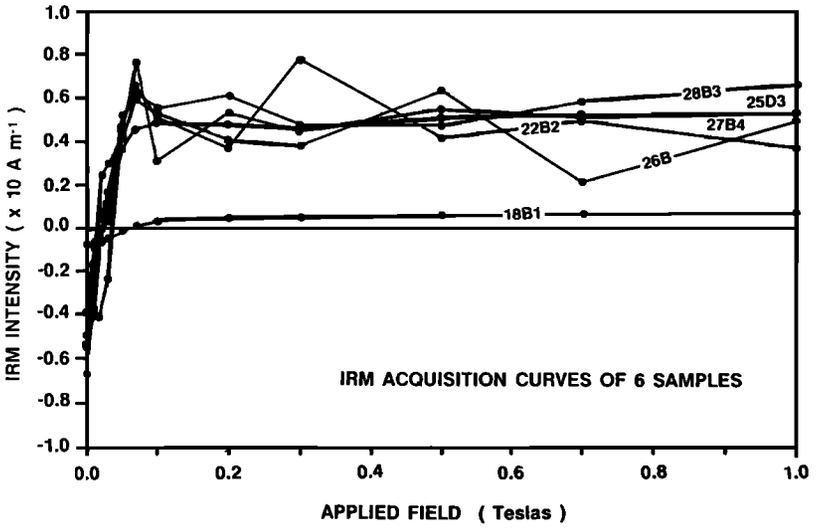


Fig. 2. Acquisition of IRM for six specimens.

intensities were measured with a cryogenic magnetometer that has a sensitivity of 10^{-6} A m^{-1}. Magnetic susceptibilities of the specimens were measured with a Bartington susceptibility bridge. All of the experiments and measurements were conducted at the Magnetics Laboratory of the University of Minnesota.

Repeated measurements of sample orientations in the field suggest an error of about 3° in azimuth. Small errors in magnetic direction and magnetic moment may have occurred during specimen preparation, owing to position of samples in the sample holder, and estimation of sample volume; multiple measurements were often taken to verify the accuracy of the results.

MAGNETIC PROPERTIES OF SPECIMENS

The magnetic results from the two granites reveal some differences in their magnetic properties. The Hong Kong granite shows a range of NRM intensity from 10 to 10^{-2} A m^{-1}; with 70% of the specimens in the 10^{-2} A m^{-1} range. A substantial range in the value of magnetic susceptibility of the Hong Kong granite is apparent. The Cheung Chau granite shows a larger range of NRM intensity, from 10^{-2} to 10^{-4} A m^{-1} and generally low susceptibility.

IRM measurements were conducted on five specimens from each granite. The specimens were first given a magnetizing field of 1 T to acquire a saturated magnetism and then an antiparallel IRM in progressively increasing fields up to 1 T. The coercivities of remanence ($\mathbf{H_{cr}}$) of the specimens were determined graphically from the IRM acquisition curves.

IRM acquisition curves for six of the specimens are depicted in Figure 2. The specimens generally acquire a magnetic saturation at about 0.1 T, indicative of a multidomain titanomagnetite as the principal magnetic carrier. The $\mathbf{H_{cr}}$ displays an inverse relationship with the magnetic susceptibility of the specimens

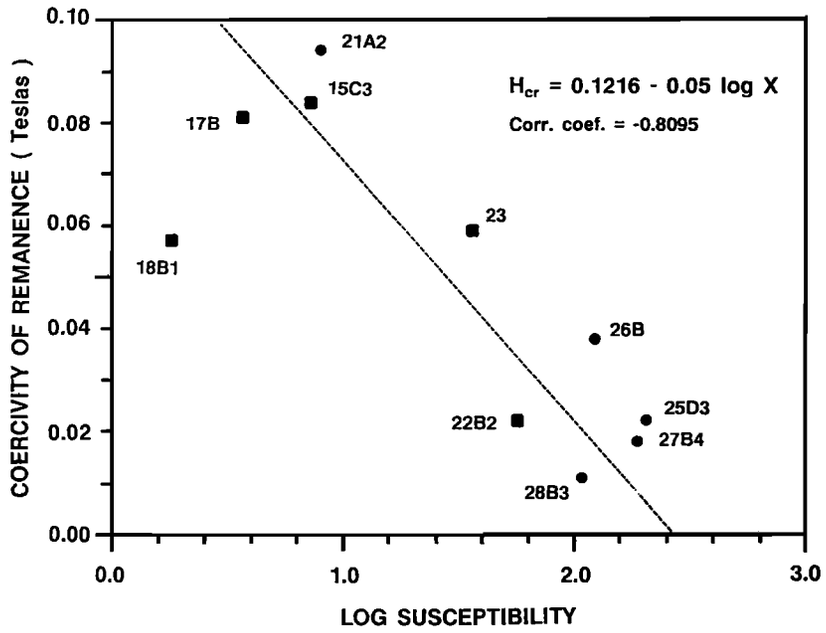


Fig. 3. Plot of coercivity of remanence against normalized susceptibility of samples. Susceptibility measurements are given in relative units.

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(Figure 3). The relation between the two magnetic parameters has been discussed by many paleomagnetists [e.g., *Stacey and Banerjee*, 1974; *O'Reilly*, 1984]. Both parameters are related to the particle size of the magnetic carriers; large magnetites probably carry a soft magnetism (low $\mathbf{H_{cr}}$) and show high magnetic susceptibility, and vice versa. Petrographic and mineralogical studies revealed more abundant and larger titanomagnetite grains in the Hong Kong granite [*Nau*, 1988; *Chan*, 1988], hence accounting for the generally higher values of magnetic susceptibility than the Cheung Chau granite.

The results from the alternating field (AF) and thermal demagnetization experiments reveal remagnetization and multiple magnetic components in the samples. Substantial variation in magnetic properties is also present among the samples studied. Some of the specimens, especially those with $\mathbf{H_{cr}}$<20 mT (200 Oe) contain unstable magnetic components.

Orthogonal projections of the demagnetization results of four specimens are shown in Figure 4. Sample HK15B represents one of the specimens effectively demagnetized by the AF method (Figure 4a). A viscous component with a northeasterly declination was removed at 10 mT. The magnetic directions above 10 mT reveal a single component. The median destruction field is 35 mT, suggesting magnetite as the magnetic carrier. The NRM of specimen HK29C2 shows two components with different polarity upon thermal demagnetization (Figure 4b). The specimen was stepwisely demagnetized up to 600°C, with 97% of the NRM intensity removed at 500°C. A reverse component appears above 550°C; this component is present in all specimens from the same site and is chosen to represent the primary direction of the specimens. The thermoremanent magnetism curve of HK18A3 shows a stable, reverse component (Figure 4c). Over 60% of the remanent magnetism was removed at 200-550°C, suggesting a titanomagnetite as the principal magnetic carrier. AF demagnetization of HK25E2 reveals a reverse component below 10 mT (Figure 4d). This component, however, is not present in rocks adjacent to a mafic dike and probably represents

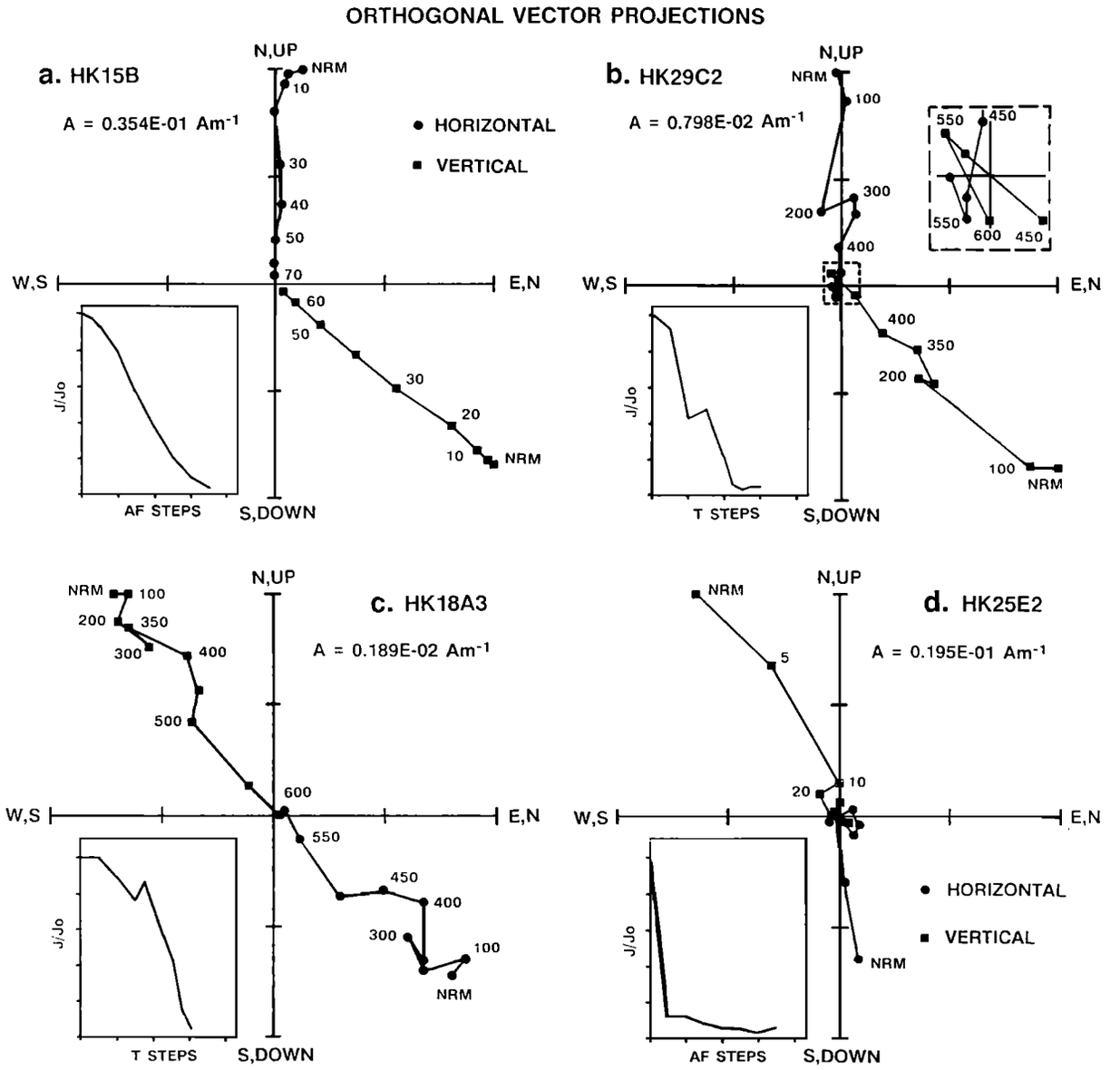


Fig. 4. Vector projections of demagnetization curves for specimens (a) HK15B (AF), (b) HK29C2 (thermal), (c) HK18A3 (thermal), and (d) HK25E2 (AF). AF steps are measured in milliteslas and temperature in °C. Normalized NRM intensities at different demagnetization steps are shown in the insets.

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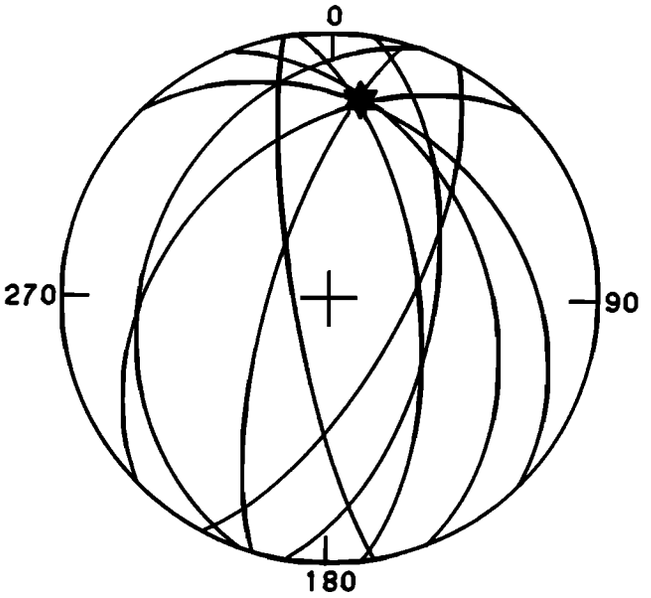


Fig. 5. Remagnetization circles of site 21. The intersection of the remagnetization circles is used as the representative direction of the site.

an acquisition before the dike intrusion, while the more coercive component above 20 mT may represent a secondary component.

In the present study, a two-pass analytical procedure was employed for determining representative directions of the samples. In the first pass, *Kirschvink's* [1980] method for vectorial analysis of magnetic components was applied for identifying a prominent magnetization in the specimens. Any set of three or more consecutive data points distributed within 10° of their mean direction is regarded as a linear segment. The linear segments obtained from all of the specimens were first grouped indiscriminantly; the most significant cluster of the linear segments represents the characteristic remanent magnetism (ChRM) of the granites.

In the second pass, the demagnetization result of individual specimens was examined for the presence of the ChRM. If a linear segment similar in direction to the ChRM was present, the linear segment was chosen as the representative direction of the sample. If no linear segment could be selected the magnetic direction of the last AF demagnetization step or of the one before the Curie temperature in thermal demagnetization was used as the representative direction of the specimen. The demagnetization result of a sample was rejected only if the sample showed unstable directions upon demagnetization. For most specimens, AF or thermal demagnetization was sufficient

TABLE 2. Paleomagnetic Results From Upper Jurassic-Creatceous Granites in Hong Kong

Site	Unit	N	Range of Representative Direction	J	I	D	α_{95}	K	P_{lat}	P_{long}	A_{95}	Polarity	Quality rating
14	CC	3	T300-OR,T500,T0-200	4E-4	-39.5	111.8	47.7	7.74	27.7	38.2	...	R	C
15	CC	5	A30-OR,A0-OR,A5-40, A10-40,A50-70	5E-3	47.0	3.2	18.2	18.56	84.0	142.4	21.5	N	A
16	CC	5	A40-OR,A20-40,A50, A60,A70	1E-4	-23.6	132.8	41.7	4.32	-44.0	199.0	28.6	R	C
17	CC	6	A50,A80,A60,A70, T0-OR,T30-50	1E-3	43.5	2.3	12.2	30.94	86 8	154.2	14.8	N	A
18	CC	9	A20,T415,A30,A80,A70,A0-40,A40,T450-550,A400-OR	2E-4	49.3	22.2	21.2	6.87	68 9	179.4	26.8	M	B
19	CC	2	A80(2)	7E-3	2.0	85.5	...	...	...	...	...	R	C
20	CC	6	A0-OR,A0-50,A0-70(2), T0-465,A20-40	1E-1	37.1	347.8	10.7	39.95	78.5	14.9	11.4	N	A
21	HK	6	Demag. Circle	5E-3	31.0	7.0	...	...	80.9	246.4	..	.	C
22	CC	5	A5-60,T0-415,A20(2),T465	3E-2	62.9	330.5	48.2	3.47	57.8	72.7	63.3	M	C
23	CC	2	A5-OR,T0-465	3E-2	51.9	340.9	...	...	...	...	...	N	B
24	CC	7	A60,A50(2),A20,A10, A40-60,T365	5E-3	44.5	42.6	36.3	3.72	51.3	190.4	40.0	M	B
25	CC	8	A5-OR(3),A0-50,A0-OR(4)	6E-2	40.3	10.4	9.3	36.24	80.5	202.0	10.3	N	A
25E	CC	3	A0-10,A70,HD	3E-2	52.6	5.9	22.6	30.80	78.6	140 0	28.6	R	B
26	HK	4	A10,T100,T210,A40	1E-3	61.9	22.1	39.8	6.29	62.9	151.1	52.8	R	B
27	HK	6	A80,A40(2),T510,A20,A50	2E-2	34.2	10.6	27.4	6.95	79.2	225 2	29.6	R	B
28	HK	7	A10,A60,A50,A0-10,A0-30, T350,T450	2E-3	50.0	55.5	25.9	6.39	40.4	182.5	32.1	R	B
29	HK	6	A5-60,T510,A40-OR,T475-OR,T115-365,T550	8E-3	42.5	7.2	17.4	15.73	83.3	188.6	21.0	M	A
All sites		17			47.5	1.3	14.6	6.9	84.3	125.8	14.6	M	
Sites A & B		12			47.8	11.5	8.7	26.1	78.2	171.9	10.6	M	

CC denotes Cheung Chau granite; HK, Hong Kong granite; J, NRM intensity in A m^{-1}; P_{lat}, latitude of paleomagnetic pole; P_{long}, longitude of paleomagnetic pole; N, normal; R, reverse; M, mixed polarity.

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to remove viscous magnetic components and secondary overprints and to reveal the primary magnetizations.

A different procedure was used for delineating the representative direction for site HK21. The specimens from this site show no consistent, stable directions, but the remagnetization circles delineated by Kirschvink's procedure of vectorial analysis intersect at a direction similar to that of the ChRM (Figure 5). For this site, the intersection vector of the remagnetization circles is regarded as the representative direction.

Since the magnetic properties of the samples are not uniform, it is necessary to rate the quality of individual site results. The quality of the data from each site is rated based on the stability of the remanent directions, the α_{95} of the site, and the efficacy of the demagnetization. Sites with an A rating generally show stable remanence, tight clustering of sample directions, and effective demagnetization of the specimens. Sites that fail one or more than one of the criteria are assigned respectively the B or C ratings. The sites with C rating generally contain dual polarity and show intermediate directions. Table 2 summarizes the demagnetization results and the fields at which the representative directions were chosen.

SUMMARY OF RESULTS

No tectonic correction or fold test was applied to the magnetic directions. The lack of orientation markers in the granites prohibits a precise determination of the rotation history of the sample sites. Mafic dikes intruding the granites, however, have a vertical orientation and show a steep principal axis of magnetic anisotropy. The field orientation and magnetic anisotropy of the dikes, therefore, suggest lack of substantial tilting movements since the Late Cretaceous-early Tertiary intrusion. The granitic bodies might have been tectonically rotated during the Cretaceous, but the similar NRM directions of different plutons separated by wide distances indicate the absence of significant tectonic rotations of individual plutons.

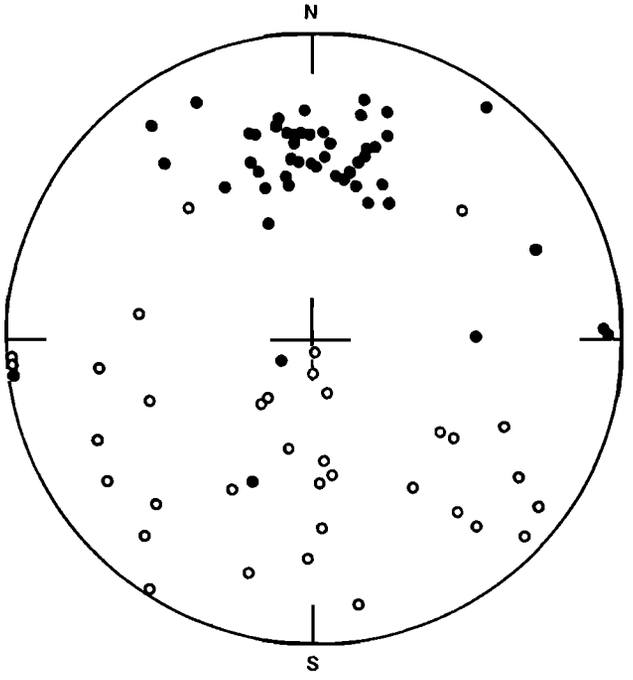


Fig. 6. Equal area projections of linear segments delineated from demagnetization results from granite samples. Open circles are data points plotted on the upper hemisphere.

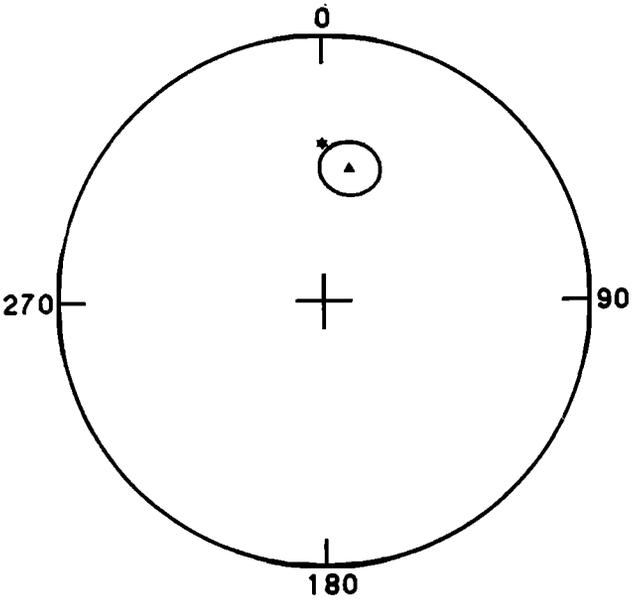


Fig. 7. Mean direction (triangle) of 16 sites with A and B quality ratings and the direction of the present field (star).

Representative directions form two antiparallel clusters on an equal area projection (Figure 6), with the reverse components showing greater dispersion than the normal components. The greater dispersion of the reverse directions is probably due to insufficient removal of magnetic overprints or significant overlap in coercivities of the magnetic carriers for different components. In this study, the paleomagnetic pole is obtained by averaging the virtual geomagnetic poles of the individual sites. The mean direction of all sites is I=47.5°, D=1.3°, and α_{95}=14.6° and yields a paleomagnetic pole at 84.3°N, 125.8°E. The mean direction of the sites with A and B ratings is I=47.8° and D=11.5° which yields a paleomagnetic pole at 78.2°N, 171.9°E, A_{95}=10.6°. The latter pole is statistically similar to the one derived from the overall mean. The better quality of the site data, however, suggests a more reliable pole based on data points that are less affected by insufficient demagnetization and overlapping coercivity spectra of multiple components. For this reason, the paleopole obtained from sites with ratings A and B is used to represent the mean paleomagnetic pole of the granites.

Although the mean direction is barely distinguishable from the present field direction (Figure 7), there are additional reasons to believe that the remanent magnetism actually represents a pre-Cenozoic acquisition. First of all, the dual polarity present in the samples implies a magnetization older than the Brunhes magnetic interval. Secondly, the remagnetization history of site HK25 has enabled a baked contact test on the time of the magnetic acquisition. The sample site in the Diamond Hill quarry in Kowloon City contains a 20 m dolerite dike. Samples were collected from the dike (sites HK25A-B), from the granite within a meter of the dike margin (HK25C-D), and at a site located about 100 m away from the dike (HK25E). All of sites HK25A-D bear a normal polarity, while site HK25E yields a reverse polarity (Figure 8). The granite probably acquired an initial reverse magnetism and was remagnetized by the dike during an interval of normal polarity. While sites 25C and 25D acquired a normal polarity during the remagnetization, site 25E was located sufficiently far away from the dike to retain a reverse polarity. Since the dolerite dike is Late Cretaceous-early Tertia-

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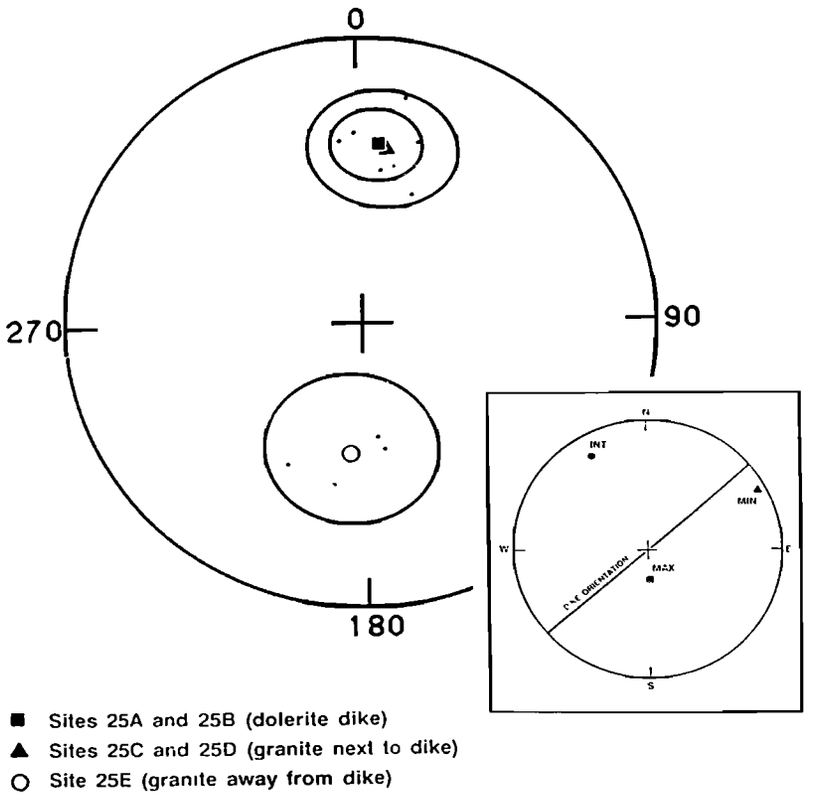


Fig. 8. Baked contact test at site HK25. The mean directions of the dike (HK25A-B) and the granite within a meter of the dike (HK25C-D) are similar and show normal polarity. A far site (HK25E) of the granite shows a reverse polarity. Inset: the vertical orientation and steep magnetic susceptibility of the dike suggest lack of substantial tilting movements of the site.

ry, the NRM acquisition must be pre-early Tertiary. Moreover, the geomagnetic field polarity was dominantly normal between 70 m.y. and 120 m.y; the reverse component in the granite, therefore, represents an acquisition either before 120 m.y. or after 70 m.y. during the Late Cretaceous. The latter interpretation is preferred over the first one because potassium-argon dating of the Hong Kong granite has shown a cooling age of 117 Ma, which lies within the Cretaceous Long Normal Magnetic Interval. Magnetic results from the granites can therefore be used to determine the Upper Cretaceous paleomagnetic pole of the study area.

TECTONIC IMPLICATIONS

Very few Upper Cretaceous paleomagnetic directions from South China are available for comparison with the results from this study (Figure 9). *Lin et al.* [1985] used the Upper Cretaceous pole from South Korea determined by *Otofuji et al.* [1983] to construct the provisional polar wander curve of South China. *Nashida et al.* [1980] reported a paleomagnetic pole from the Cretaceous Port Island formation in Hong Kong; later results from the same formation, however, fail the fold test and suggest a Tertiary remagnetization [*Chan and Zich*, 1986]. *Kent et al.* [1986] reported two paleomagnetic poles from Upper Cretaceous-Tertiary formations in east-central China and west Sichuan. The paleomagnetic pole position (Latitude, 76.3°N; Longitude, 172.6°E; $A_{95}=10.3°$) from east-central China is statistically indistinguishable from that reported in this paper. The paleopole from west Sichuan, however, reveals a significant relative motion between east China and west Sichuan; this relative motion probably resulted from the India-Eurasia collision. A paleomagnetic pole position (66.0°N, 221.5°) of the Upper Cretaceous red basins in Guangdong Province in South China also differs significantly from the pole reported in this study [*Hsu*, 1987], but there are no particular reasons to exclude this result from the computation of the mean pole. Excluding the two paleomagnetic poles from the Port Island formation that fail the fold test, and the west Sichuan site that shows an apparent relative motion, the paleomagnetic poles from east China, South Korea, Guangdong, and Hong Kong yield a mean pole at 73.5°N latitude, 194.3°E longitude, $A_{95}=10.4°$.

The paleomagnetic pole of South China is statistically indistinguishable from the Eurasia pole [*Irving*, 1977] and the North China pole [*Lin et al.*, 1985] (Figure 9). The three continents, therefore, have not experienced significant relative latitudinal movement since the Late Cretaceous. Small latitudinal movements under 1000 km, however, cannot be resolved with the available paleomagnetic data. Although the three paleopoles are located at about the same latitudes, all previous studies have revealed consistently a separation of the Cretaceous paleomagnetic poles of South China from the Eurasia pole by about 20-30° along the 74°N latitude. *Lin et al.* [1985] suggested that this separation was caused by a rotation of South China relative to North China about an axis near the North Pole. This sinistral motion between the two blocks probably resulted from tectonic extrusions during the Cenozoic India-Eurasia collision [*Tapponnier et al.*, 1982]. *Courtillot and Besse* [1986] suggested that the offset between the Lower Cretaceous poles of Eurasia and South China could have resulted from sinistral shear along the NNE--SSW Tanlu fault in eastern China about a pole located about 90° from and at right angles to the fault. The Upper Cretaceous

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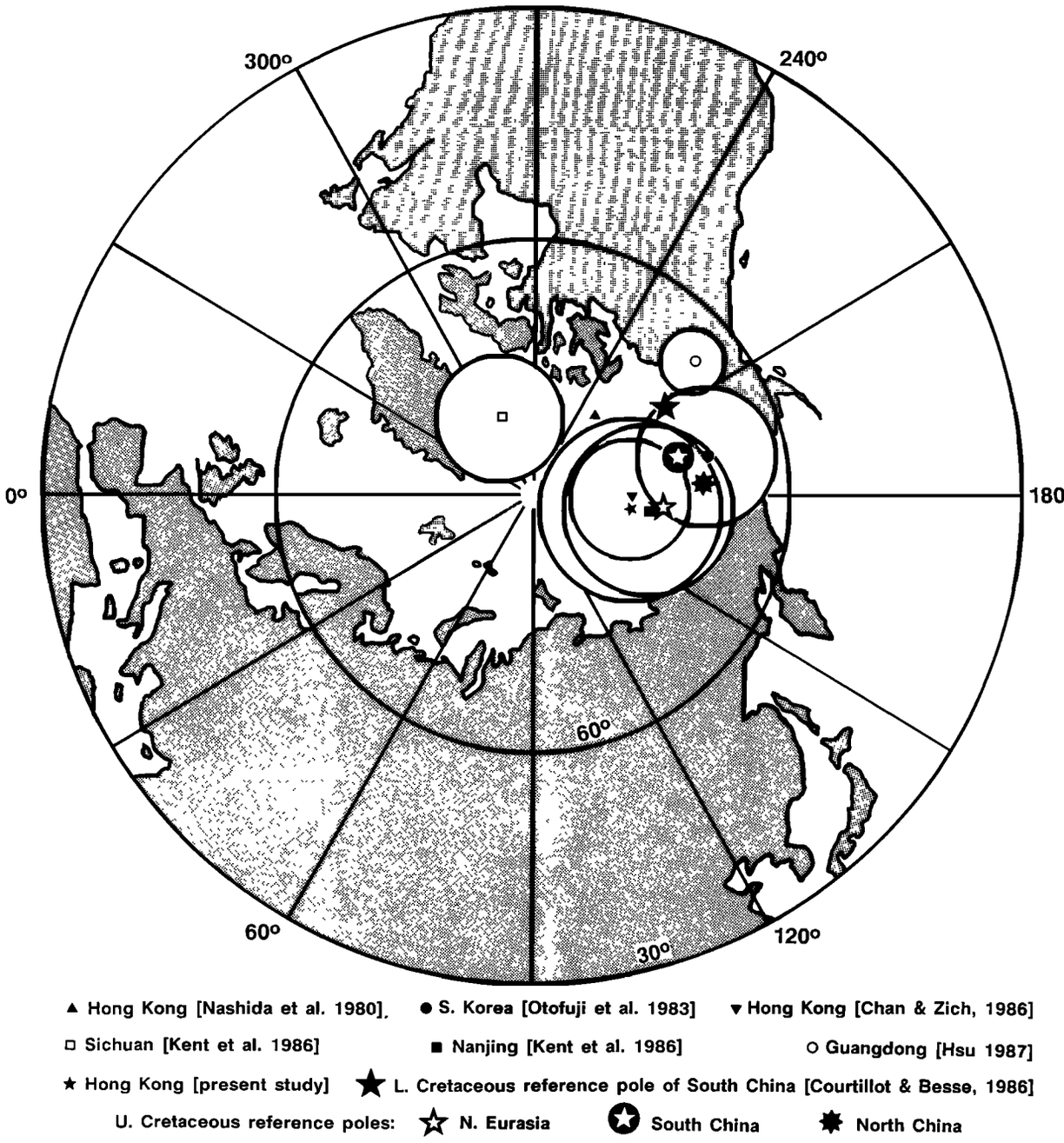


Fig. 9. Cretaceous paleomagnetic poles of South China. Small symbols are paleomagnetic poles of South China reported by previous studies. The larger symbols are the Upper Cretaceous reference poles for North Eurasia [*Irving*, 1977], North China [*Lin et al.*, 1985], and South China [this study], and the Upper Jurassic-Lower Cretaceous pole of South China [*Courtillot and Besse*, 1986].

reference pole of South China appears to lie halfway between the Lower Cretaceous pole of South China (73°N latitude, 212° longitude [*Courtillot and Besse*, 1986]) and the Cretaceous reference pole of North Eurasia (75°N latitude, 170° longitude [*Irving*, 1977]) which occupied a fairly stable position during the Late Cretaceous. The spatial relation among the three poles probably suggests that the relative movement between South China and Eurasia might have begun during the Late Cretaceous. The proximity of the paleomagnetic poles to each other and their intersecting domains of confidence, unfortunately, have hindered the determination of a unique Euler pole for the relative motion. A more precise location of the Euler pole can be determined by comparing the entire Phanerozoic polar wander paths of South China and Eurasia. But the current precision and accuracy of the paleomagnetic data from South China have rendered such comparison possible. The current tectonic interpretation of the available paleomagnetic data from China is tentative; the immediate concern of paleomagnetic studies in China should focus on acquisition of reliable paleomagnetic data from various parts of China.

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